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Research Paper

Point transformer-based 3D segmentation and phenotypic trait analysis of cotton plants with foliage

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ABSTRACT

High-resolution point clouds provide detailed three-dimensional (3D) spatial information about cotton plants, making them valuable for analyzing complex architectural traits. However, segmentation of cotton bolls and other plant components under dense foliage remains underexplored, although accurately measuring boll-related and architectural traits before defoliation reflects cotton's natural growth conditions and is valuable for breeders. The goal of this study was to leverage a state-of-the-art transformer-based point cloud segmentation model, Point Transformer V3 (PTv3), to segment cotton plants with dense foliage into distinct organs and derive architectural traits. PTv3 was compared with three other models: PointNet++, PointNeXt, and PTv2, to segment cotton plants into distinct organs with up to four classes on two datasets with different levels of challenge. Post-processing methods were developed to count bolls and derive other architectural traits. The experimental results showed that PTv3 achieved the best segmentation performance across all tested segmentation scenarios. The mean absolute percentage error (MAPE) for cotton boll counting, main stem length, main stem diameter, branch angle, and internode distance were 9.55%, 4.09%, 9.23%, 7.04%, and 14.16%, respectively. Moreover, the spatial distribution of cotton bolls and foliage along the vertical direction in different genotypes was successfully obtained and analyzed. This method provides a promising approach for analyzing leafy cotton plants by enabling accurate extraction of structural and yield-related traits under natural growth conditions. The phenotyping information supports the evaluation of architectural differences prior to harvest, thereby advancing cotton phenotyping research and improving breeding efficiency.

1. Introduction

As the global population continues to grow, developing effective strategies to enhance crop yields presents a significant challenge in modern agriculture (Voss-Fels et al., 2019). Cotton (*Gossypium hirsutum* L.), one of the most widely cultivated economic crops, contributes approximately $50 billion to the global raw cotton market (Devoto et al., 2024). Improving cotton yield requires optimizing various agronomic factors, with the architectural traits of cotton plants playing a crucial role. These traits affect light interception, planting density, mechanized harvesting efficiency, and breeding strategies, ultimately influencing productivity and economic returns (Li et al., 2024; Reinhardt and Kuhlemeier, 2002). Traditionally, plant architectural traits have been assessed through manual measurements, which are time-consuming, labor-intensive, and often limited to small sample sizes, restricting large-scale phenotyping efforts (Furbank and Tester, 2011). Additionally, 2D image-based analysis has been widely used to improve efficiency, but its ability to phenotype plant structures is limited by occlusion, restricting the analysis of complex traits (Kenchanmane Raju et al., 2020). These limitations hinder precise phenotypic analysis and reduce the reliability of trait measurements (Kolhar and Jagtap, 2023). In contrast, three-dimensional (3D) point cloud data provide precise geometric and spatial information, making them particularly advantageous for analyzing complex plant structures (Harandi et al., 2023). In cotton plants, point clouds typically consist of foliage, cotton bolls, main stems, and branches, and accurately segmenting these components is crucial for structural analysis and trait extraction.

In recent years, point cloud data has been increasingly utilized for plant phenotypic analysis, particularly in the 3D structural assessment of various plants. In crops such as maize and sorghum, where foliage remains independent and do not overlap within a single plant, point cloud

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skeletonization algorithms have been employed to segment the main stem and foliage. This has facilitated the extraction of key traits, including stem length, stem diameter, leaf length, and leaf angle (Miao et al., 2021; Xiang et al., 2019). Traditional machine learning techniques, such as support vector machines (SVM) and random forests (RF), rely heavily on manually crafted feature extraction, limiting their ability to generalize to more complex or diverse plant structures (Dutagaci et al., 2020; Sodhi et al., 2017; Zhou et al., 2019; Ziamtsov and Navlakha, 2019). Additionally, these methods are effective for basic classification and segmentation tasks, but their generalization ability deteriorates when applied to intricate plant architectures. In contrast, deep learning techniques, such as PointNet++ (Qi et al., 2017b), have automatically learned features directly from raw point cloud data, significantly improving segmentation accuracy and robustness. PointNet++ and its improved models have been successfully applied to various plants, including sorghum (Patel et al., 2023), and maize (Yang et al., 2024). Although PointNet++, one of the earliest deep learning-based point cloud segmentation models, has achieved wide adoption in the segmentation of various crops, most of its applications have been limited to plants with relatively simple architectures. For instance, maize plants primarily consist of stems and foliage, with distinct structural differences that facilitate segmentation. Moreover, some studies have addressed the segmentation of crops from the ground surface, which inherently reduces the complexity of the task (Shi et al., 2024). These studies demonstrate the feasibility of using deep learning for plant point cloud organ segmentation and trait extraction. Nevertheless, the segmentation and analysis of more structurally complex plants remain an open area for further investigation.

Point cloud segmentation based on deep learning has been applied to cotton plant organ segmentation. The PEPNet point cloud segmentation network was designed for stem and leaf segmentation of cotton seedlings, achieving an mIoU of 91.31% (Yan et al., 2024). The seedling stage is a critical phase in cotton growth, significantly impacting subsequent development and yield (Shen et al., 2024). As cotton plants grow, their foliage begins to overlap, significantly increasing the difficulty of segmentation. The PointSegAt model was used to segment cotton plant stems and foliage while distinguishing leaf overlap (Hao et al., 2024). Leaf occlusion often leads to information loss, necessitating additional point cloud completion work to restore the occluded areas (Li et al., 2023). In these studies, the plants were cultivated in pots inside greenhouses, and the data acquisition was carried out under controlled indoor conditions. In a few other studies of cotton plants during the boll formation or maturation stages, models such as PointNet++ and PVCNN have been used to achieve boll segmentation. However, the segmentation of the main stem and branches requires post-processing to meet the desired results (Saeed et al., 2023). In these studies, the cotton plants were defoliated either by chemicals or manually to reduce occlusion, significantly alleviating segmentation difficulties. CottonSense (Bolouri et al., 2024) effectively segments and counts cotton fruits at four growth stages—square, flower, closed boll, and open boll—thereby overcoming the challenges of deployment across multiple growth periods. However, this approach primarily concentrated on cotton bolls, without conducting an in-depth analysis of other plant organs. In cotton plants, a comprehensive analysis of the structural characteristics of the main stem, branches, and foliage, alongside cotton bolls, during the boll formation and maturation stages would offer valuable insights into plant development and growth dynamics.

Segmenting cotton plants with foliage is particularly challenging yet crucial for practical field phenotyping, as leaves remain throughout most of the growth season and only shed naturally at boll maturity. In breeding programs, cotton plants are typically analyzed during their natural growth phases, when foliage is still present (Zhang et al., 2013). Under these conditions, accurate 3D segmentation enables non-destructive quantification of key traits such as boll count, canopy structure, and branch angles—traits closely linked to yield potential and resource use efficiency (Zhai et al., 2024). Over the years, point cloud segmentation models have evolved significantly, leading to notable improvements in both accuracy and efficiency, especially on challenging datasets. The development began with the original PointNet (Qi et al., 2017a), which introduced deep learning for unordered point sets, and was followed by PointNet++, which enhanced this approach through hierarchical learning. PointNeXt (Qian et al., 2022) introduced architectural optimizations based on PointNet++, further improving computational efficiency and scalability, making it more suitable for large-scale point cloud processing. Since the introduction of the Vision Transformer (ViT, Dosovitskiy et al., 2021), transformer-based models have made significant progress in computer vision. By utilizing self-attention to capture global information, transformers excel at handling long-range dependencies and complex scenes (Ramachandran et al., 2019; Vaswani et al., 2017). Their success has also spurred the adoption of transformer architecture in the 3D domain. Point Transformer (Zhao et al., 2021) was the first to introduce the transformer into the point cloud domain, followed by several improved versions, such as Point Transformer V2 (PTv2, Wu et al., 2022) and Point Transformer V3 (PTv3, Wu et al., 2024). PTv3 achieved state-of-the-art performance on public datasets for point cloud semantic segmentation. These advancements in point cloud segmentation provide a strong foundation for accurately segmenting cotton plants with foliage, addressing challenges such as occlusions, complex plant structures, and fine-grained object separation.

In this study, a transformer-based point cloud segmentation network was introduced to segment cotton plants with dense foliage. These cotton plants were in the boll maturation stage, where most of the bolls remained green and were surrounded by dense foliage, which had not been fully investigated before. We selected the state-of-the-art Point Transformer V3 (PTv3) for segmentation and compared its performance with other models. This experiment not only segmented cotton bolls but also successfully distinguished branches, the main stem, and foliage, enabling further analysis of different plant organs. The overall goal of this study was to develop a 3D point cloud processing workflow for phenotypic trait extraction of cotton plants with foliage using terrestrial LiDAR data. Specific objectives were to: (1) compare different deep learning models in cotton boll segmentation with foliage; (2) evaluate the segmentation performance of PTv3 in two-class, three-class, and four-class classification tasks; (3) conduct a comparative evaluation of model performance on two datasets with different levels of structural complexity; (4) propose a post-processing method to improve cotton boll counting accuracy and to extract multiple architectural traits of cotton plants.

2. Materials and methods

2.1. Data collection

This experiment primarily focuses on the study of individual cotton plants with foliage. The cotton plants were obtained in November 2021 from a breeding field at Iron Horse Farm in Greene County (latitude: 37.730 N, longitude: 83.303 W), Georgia, USA. A total of 20 individual cotton plants were collected from the field, cut at the soil surface, and then transferred indoors. At this stage, most cotton bolls were green and unopened. The plants were secured with a support frame during data collection, ensuring no overlap between them. A high-resolution 3D LiDAR scanner (FARO Focus S70, FARO Technologies, USA) mounted on a tripod at a height of about 0.6 m was used to scan the cotton plants from multiple locations over a 360° horizontal plane (Fig. 1a). Color information was captured using the built-in color camera. Scans from multiple angles were aligned and merged into a single point cloud using FARO SCENE software (version 2019.2, FARO Technologies, Florida, USA). The self-collected cotton plant point cloud dataset, characterized by its structural complexity, is referred to as CottonStruct-H (H for high complexity) in this study. To compensate for the limited number of collected samples, an additional 80 publicly available cotton

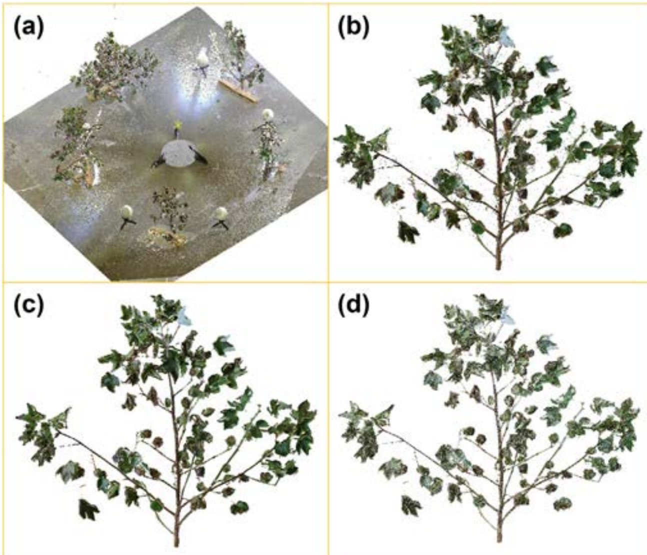

Fig. 1. Data collection and preprocessing workflow. (a) Experimental setup and raw point cloud acquired using terrestrial LiDAR. (b) An individual cotton plant with foliage. (c) Point cloud denoising. (d) Point cloud down-sampled to 100,000 points.

point cloud samples with relatively simple structures were used (Liu et al., 2025), collectively referred to as CottonStruct-L (L for low complexity). These two datasets were used independently for training and comparative analysis of segmentation performance. The point clouds in CottonStruct-H contain coordinate and color information, whereas those in CottonStruct-L contain coordinates and normals.

2.2. Data preprocessing and annotation

The pre-processing of the point cloud was conducted to isolate individual cotton plants with foliage. First, the registered point cloud was imported into the open-source software CloudCompare. A single plant was manually cropped from the point cloud, and non-plant elements, such as the ground and bracket points, were removed (Fig. 1b). To eliminate noise, a statistical outlier removal (SOR) filter was applied. This filter estimated the mean distance of each point to its six nearest neighbors and removed points whose distances exceeded one standard deviation (Fig. 1c). Finally, spatial down-sampling in CloudCompare was used to reduce the point cloud of each plant to 100,000 points (Fig. 1d).

The pre-processed individual cotton plants with foliage were labeled into different classes, and saved in S3DIS format (Armeni et al., 2016). In this experiment, three different classification schemes were applied: 2 classes (cotton bolls and non-cotton bolls, Fig. 2a), 3 classes (cotton bolls, main stems, and others, Fig. 2b), and 4 classes (cotton bolls, main stems, branches, and foliage, Fig. 2c) to evaluate the impact of different annotations on segmentation. All 20 plants were annotated. The CottonStruct-H dataset was divided into four groups for cross-validation, with each group containing five plants. In each cross-validation iteration, three groups were used for training, and one group was used for testing. The CottonStruct-L dataset was also annotated with three fine-grained semantic classes (Fig. 2d, e, f). Each point cloud was uniformly downsampled to 40,960 points. A total of 50 samples were used for training and 30 for testing, without cross-validation.

The point cloud composition of different cotton plants was analyzed. In the CottonStruct-H dataset, the horizontal stacked bar chart illustrates the distribution of four labeled categories: foliage (blue), cotton bolls (orange), main stems (yellow), and branches (purple) across individual cotton plants (Fig. 3). Each bar represents a single plant, with category proportions normalized to the total number of points in that plant. The results indicate that foliage dominates the point cloud in most plants, followed by cotton bolls, while the proportions of main stems and branches vary across samples. The relatively low proportions of stems and branches can be attributed to heavy occlusion by dense foliage. The selected genotypes of the plants are listed in Table 1. In the CottonStruct-L dataset, the proportion of branches increases significantly, which can be attributed to the sparser foliage and reduced occlusion in this dataset (Table 2). This distribution analysis provides insights into the dataset composition and aids in evaluating segmentation performance under different classification schemes.

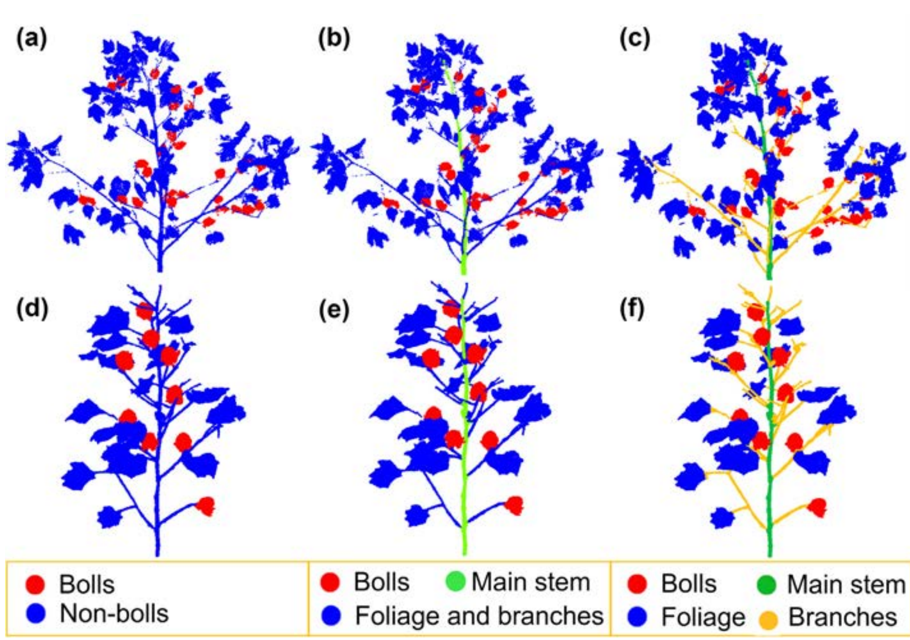

Fig. 2. Point cloud annotations with different class granularities. The top row (CottonStruct-H) presents more complex plant architecture, while the bottom row (CottonStruct-L) shows relatively simpler structures. (a, d) Two-class annotations: cotton bolls and non-cotton bolls. (b, e) Three classes: cotton bolls, main stem, and others. (c, f) Four classes: cotton bolls, main stem, branches, and foliage.

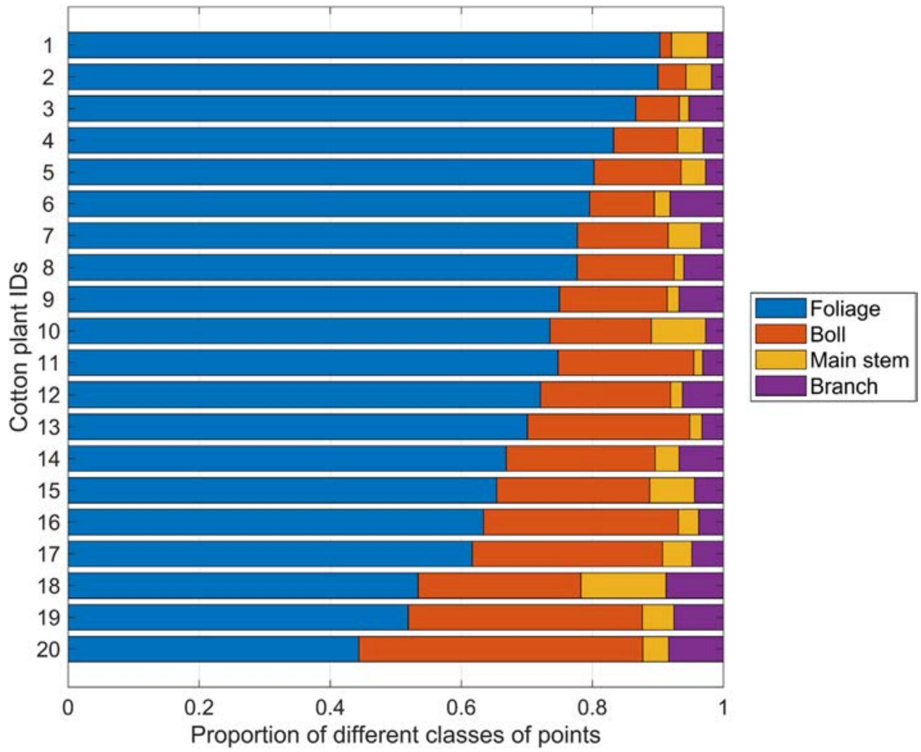

Fig. 3. Distribution of point cloud data across organ classes in cotton plants with foliage in the CottonStruct-H dataset, normalized by the total point count per plant.

2.3. Comparison of point cloud segmentation models

Point Transformer V3 (PTv3) is an advanced transformer-based architecture specifically designed for point cloud processing (Wu et al., 2024). It introduces a novel approach that prioritizes simplicity and efficiency over intricate designs, challenging the traditional accuracy-speed trade-off. PTv3 adopts an encoder–decoder framework to achieve large-scale point cloud semantic segmentation through efficient point cloud serialization and local attention mechanisms. First, the input point cloud is processed by serialization and feature embedding, converting the unordered point set into an ordered feature sequence suitable for Transformer-based processing. The encoder consists of multiple Point Transformer blocks and progressively downsamples the point cloud via grid pooling, capturing multi-scale contextual features. During decoding, the decoder restores the point cloud resolution through grid unpooling and incorporates skip connections to fuse features from different encoder stages, thereby enhancing spatial detail representation. Finally, the decoder produces a high-dimensional feature vector for each point, which is mapped to per-point class logits by a lightweight MLP prediction head, resulting in point-wise semantic segmentation.

In the PTv3 architecture, the point cloud is first initialized through two main modules (Fig. 4a): Serialization and Embedding. The Serialization module converts unordered and unstructured point cloud data into a structured format by mapping 3D points into a one-dimensional ordered sequence using specific space-filling curves, such as Z-order and Hilbert curves. Subsequently, the Embedding module encodes the serialized point cloud and projects it into a high-dimensional feature space using a multilayer perceptron (MLP), which serves as the input to the transformer encoder.

As shown in Fig. 4b, the Grid Pooling module serves as a key component in the PTv3 architecture, performing spatial partitioning of the point cloud into uniform grids and applying pooling operations within each grid. This enables effective downsampling between the encoder stages and facilitates local feature aggregation while expanding the receptive field. To mitigate overfitting to a single serialization pattern, PTv3 introduces a Shuffle Order strategy within the encoder, which randomly alters the serialization patterns used across consecutive attention layers. The encoder itself consists of multiple stacked, identical blocks. The enhanced Conditional Positional Encoding (xCPE) is a positional encoding mechanism that incorporates a sparse convolutional layer with a skip connection, placed prior to the attention layer. This design enables more efficient encoding of point positional information and serves as a substitute for the less efficient Relative Positional Encoding (RPE). Layer Normalization (LayerNorm) is employed to normalize the activations within each network layer, thereby improving training

Table 1
Genotype information of the tested plants in the CottonStruct-H dataset.

Genotypes	Number of plants
Acala Maxxa	1
DES56	2
DG3615	2
DP1646	4
DP341	2
MDN0101	2
ST5020	2
TO246BC3MDN	2
Tamcot Sphinx	2
UA48	1

Table 2
Distribution of classes in cotton plant point clouds across different datasets (the numbers in the table are in percentage %).

	Foliage	Boll	Main stem	Branch
CottonStruct-H	72.1	19.3	5.0	3.6
CottonStruct-L	65.3	17.1	5.1	12.5

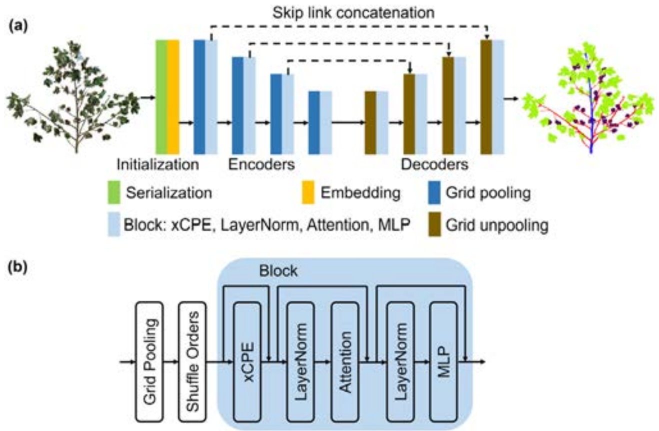

Fig. 4. Overall architecture of the PTv3 model. (a) The network follows an encoder–decoder framework. The encoder progressively downsamples the point cloud using grid pooling and extracts multi-scale features through Point Transformer blocks. The decoder restores the point resolution via grid unpooling and integrates skip connections to fuse features from different encoder stages. (b) Structure of a Point Transformer block, which consists of enhanced conditional positional encoding (xCPE), attention, normalization, and MLP layers.

stability and enhancing the model's generalization capability. PTv3 adopts a window-based dot-product attention mechanism, enabled by the structured format resulting from point cloud serialization. To further enlarge the receptive field, the model utilizes Patch Attention, which partitions the serialized point cloud into multiple non-overlapping patches and applies attention computation within each patch. Finally, the output point cloud features are generated through the MLP.

The architecture of PTv3 follows the classic U-Net (Ronneberger et al., 2015) structure, in which a four-stage encoder progressively extracts high-level features from the point cloud while reducing spatial resolution, and a four-stage decoder subsequently restores the resolution to produce the final predictions. The model comprises four encoder and four decoder stages, with block depths of [2, 2, 6, 2] and [1, 1, 1, 1], respectively. For these stages, the grid size multipliers are set to [×2, ×2, ×2, ×2], representing the expansion ratio relative to the preceding pooling stage. PTv3 expands the receptive field to 1024 points, allowing the model to capture long-range dependencies between distant points more effectively. In this study, the input point cloud data includes spatial coordinates (x, y, z) and color information (R, G, B). The batch size is set to 2, and the voxel size is set to 0.005. All other parameters are configured according to the default settings provided in the official codebase.

PointNet++ (Qi et al., 2017b) adopts a hierarchical learning strategy to extract rich features from local to global contexts in point clouds. It utilizes two core modules—Set Abstraction (SA) and Feature Propagation (FP)—to implement downsampling and upsampling, respectively. The SA module integrates sampling, grouping, and PointNet operations to enable point subsampling, neighborhood feature aggregation, and hierarchical representation learning. By stacking multiple SA layers, PointNet++ effectively captures features at varying spatial resolutions. The FP module transfers high-level features back to lower-resolution points, enabling multi-scale feature fusion. A final global pooling operation aggregates local features into a global descriptor, supporting downstream tasks such as classification and segmentation. Unlike PointNet (Qi et al., 2017a), which treats the point cloud as a whole, PointNet++ explicitly encodes local geometric structures through hierarchical feature learning, yielding improved performance in tasks requiring fine-grained spatial understanding. In our experiments, each input point cloud contained 100,000 points. The Multi-Scale Grouping (MSG) strategy was adopted for local feature extraction during the training of cotton plant point clouds. All remaining parameters followed the default settings of the official implementation.

PointNeXt (Qian et al., 2022) enhances point cloud segmentation performance through architectural improvements and the integration of Inverted Residual Multi-Layer Perceptron (InvResMLP) blocks. The network begins with an MLP layer that projects input point coordinates into a higher-dimensional feature space, followed by a sequence of Set Abstraction (SA) modules, each paired with an InvResMLP block. InvResMLP introduces residual connections to address vanishing gradients and adopts separable MLPs to reduce computational overhead while preserving point-wise feature learning. An inverted bottleneck design is also incorporated to further improve feature representation. The model's scalability in terms of depth and width is achieved via scaling strategies applied to InvResMLP layers. After feature extraction, multi-level features are upsampled and merged via Feature Propagation, which interpolates features from coarse to fine resolutions and fuses them with earlier feature maps, thereby restoring spatial details lost during downsampling. This design maintains both local geometric detail and global contextual consistency in the final prediction. In our experiments, each point cloud sample contains 100,000 points, and the batch size is set to 8. The feature channel width in each network layer is fixed at 32, and the initial radius for ball query is set to 0.01. All other parameters follow the default settings provided in the official implementation.

Point Transformer V2 (PTv2) introduces several architectural enhancements, including Grouped Vector Attention (GVA), Position Encoding Multiplier (PEM), and Partition-based Pooling, aimed at improving the efficiency and expressiveness of point cloud feature learning (Wu et al., 2022). GVA is a novel attention mechanism that partitions the channels of value vectors into multiple groups, with shared attention weights applied within each group. This grouped weight sharing strategy significantly reduces parameter count and enhances both computational efficiency and model generalization. PEM further augments the attention mechanism by applying a learned multiplier to relational vectors based on relative positional encodings, thereby enabling more effective modeling of complex spatial relationships. The combination of GVA and PEM synergistically strengthens the model's capacity for geometric reasoning. Partition-based Pooling divides the point cloud into non-overlapping spatial regions, within which point features and coordinates are directly aggregated. This operation supports spatially aligned, efficient feature extraction, and is particularly well-suited for hierarchical processing. During training, the voxel size is set to 0.001. Grid sizes for Partition-based Pooling are configured as [0.03, 0.04, 0.06] across network layers to support multi-scale feature extraction. Smaller grid sizes capture fine-grained local structures, while larger ones encode broader contextual information. This multi-scale design enhances the model's ability to capture both detailed geometry and global scene layout from point clouds.

2.4. Phenotypic trait extraction

2.4.1. Cotton boll counting

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) was used for boll instance segmentation. This method was a density-based, unsupervised clustering algorithm that identified clusters of arbitrary shapes while effectively detecting noise points. It defined clusters using two parameters: epsilon (ϵ), the neighborhood radius, and MinPts, the minimum number of points required within the ϵ-neighborhood for a point to be considered a core point. A core point, along with its neighboring core and border points, formed a cluster. Unlike many clustering algorithms, DBSCAN did not require specifying the number of clusters in advance—only these two parameters needed to be set. After multiple trials, ϵ was set to 3.5 cm and MinPts to 10, based on the density of the boll point cloud in this experiment.

After DBSCAN clustering, the post-processing method was applied to separate multiple bolls that were clustered into one cluster (Fig. 5). Each cluster was enclosed by an Oriented Bounding Box (OBB), and the OBB volume was calculated to represent the cluster's size. A volume threshold, α, was set to determine which clusters required further optimization. The volumes of the boll clusters on each plant were sorted (Fig. 5b). If a cluster's volume exceeded the maximum of 2β (twice the cluster median volume β) and 200 cm^3 (i.e., $\alpha = \max(2\beta, 200)$), the cluster required further optimization.

There was a certain gap between the two bolls, where the number of points significantly decreased. This observation was used to separate multiple bolls that had been clustered together. First, the point cloud was transformed into a new coordinate system by aligning the direction of the longest edge of the OBB with the x-axis. Then, the point cloud was segmented along the x-axis into 60 three-dimensional (3D) boxes, and the number of points within each box was counted (Fig. 5c, d). The histogram of the number of points in each box was plotted, and the local minimum of the histogram was identified (Fig. 5e).

A local minimum was determined for every 10 bins, and the boll segmentation points were then selected from these local minima. First, the two boundary local minima (i.e., those corresponding to the maximum and minimum x-coordinate values) were removed. For each of the remaining local minima (red dots in Fig. 5e), the ratio with its adjacent local minimum was calculated; if the ratio was less than 0.5, that point was considered a boll segmentation point. Additionally, if the final selection resulted in multiple adjacent boll segmentation points among the local minima, only one was randomly chosen as the segmentation point. In Fig. 5e, the red dots marked with an asterisk (*) represent the boll segmentation points. At the segmentation points, a 3D plane was constructed, which divides the cluster into individual bolls (Fig. 5f, g).

The vertical distribution illustrates the position of each cotton boll along the plant's height. To analyze the distribution of cotton bolls at different height levels, the plant was divided into vertical sections, each 10 cm thick. Since most plants range in height from 1.0 to 2.0 m, this resolution effectively captures the spatial variation of cotton bolls. The number of centroids representing cotton bolls within each section was then counted, enabling the creation of a height distribution map with a 10 cm resolution. This map provides valuable insights into the concentration of cotton bolls at various heights within the plant (Dube et al., 2020).

2.4.2. Cotton plant architectural traits

2.4.2.1. Main stem traits. The length and diameter of the main stem were estimated from the segmented main stem point cloud. The highest point A, the lowest point C, and the midpoint B of the main stem were selected (Fig. 6a). The Euclidean distances between adjacent points were calculated, and the sum of these distances ($d_1 + d_2$) was used to represent the main stem length. Due to the influence of point cloud segmentation accuracy, this study checked whether the difference between the maximum z-coordinate of the segmented main stem (i.e., the z-coordinate of point A in Fig. 6a) and the maximum z-coordinate of the entire plant point cloud exceeded 20 cm. If the difference was greater than 20 cm, the highest point of the plant point cloud was used to replace point A for main stem length estimation. The bottom 1 cm of the segmented main stem point cloud was used to estimate its diameter. These points were projected onto the XOY plane, and the projected points were fitted to a circle using the least squares method (Fig. 6b). The diameter of the fitted circle was then used as the estimated main stem diameter.

2.4.2.2. Branch traits. The angle between the branch and the main stem was calculated by fitting the direction vectors of the main stem and the branch using point cloud data. The direction vector of the main stem was denoted as $\overrightarrow{V_s}$ and the direction vector of the branch was denoted as $\overrightarrow{V_b}$ The angle θ was then defined as Eq. 1. To obtain the direction vectors of the branch and the main stem, it was first necessary to identify individual branch points and the nearby main stem points. This process involved several steps: determining the main stem for further processing, identifying the branch, extracting individual branches, locating the main stem points near each branch, and calculating the angle.

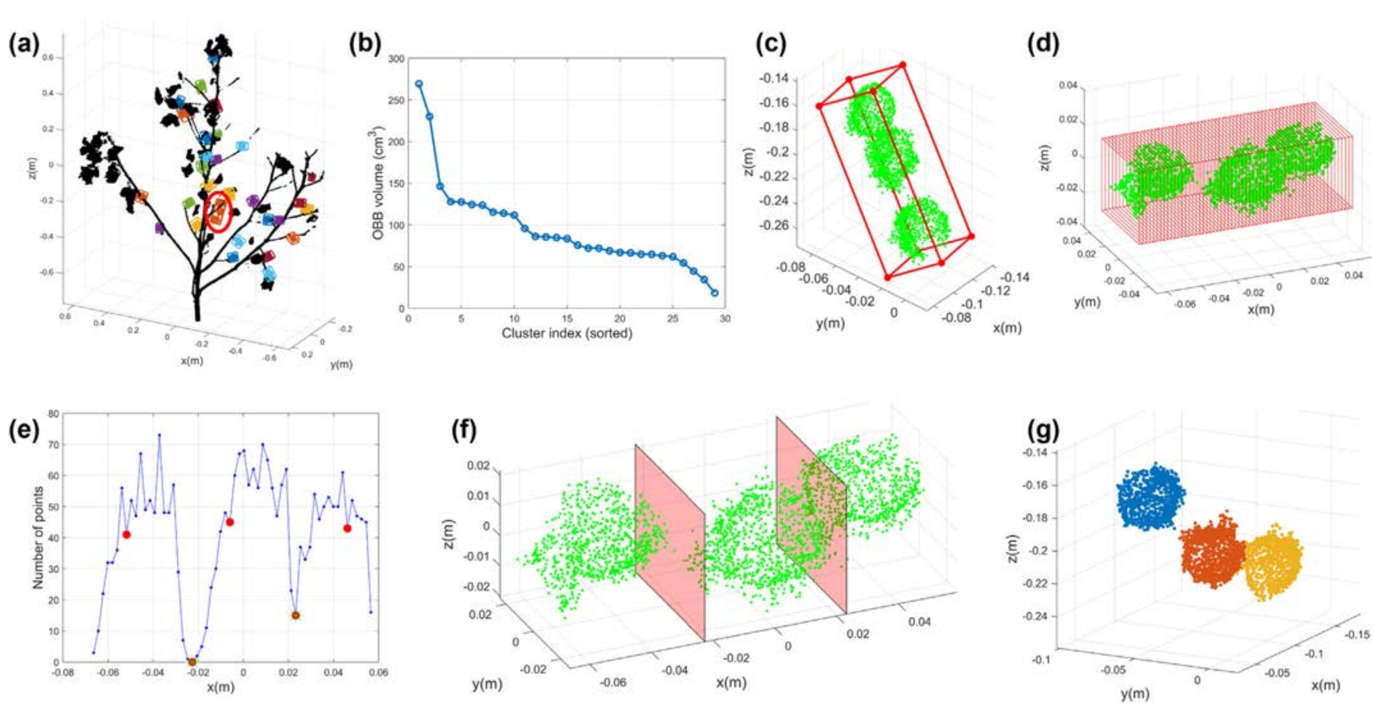

Fig. 5. An example of cotton boll clustering and post-processing. (a) DBSCAN clustering results, with each cluster assigned an Oriented Bounding Box (OBB). (b) Sorting clusters by volume. (c) An example of an OBB for a cluster with three bolls. (d) Point cloud coordinate system transformation and OBB segmentation. (e) Determining the segmentation points of the cluster. (f) Constructing the segmentation plane for the cluster. (g) Final cotton boll instance segmentation results.

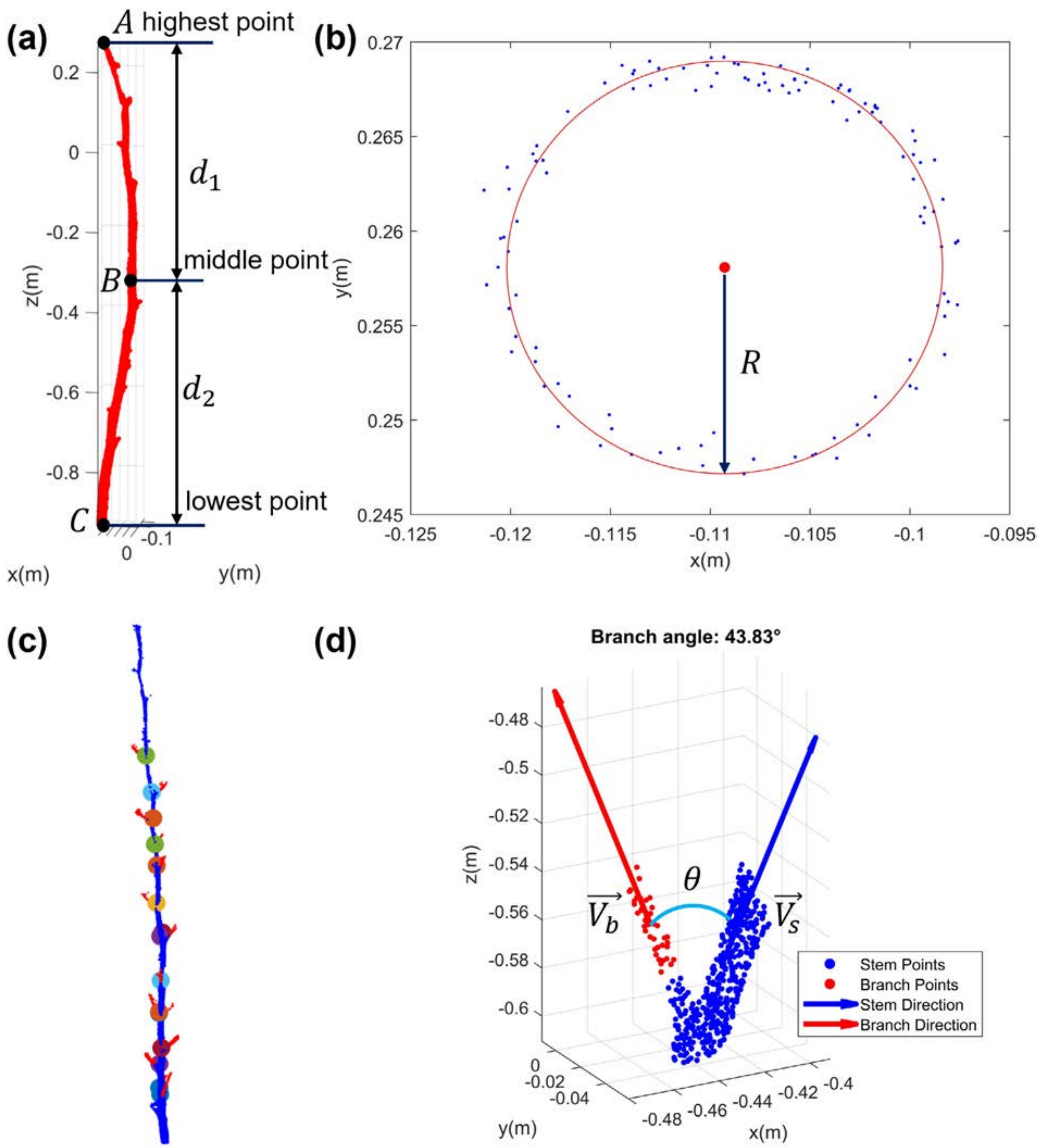

Fig. 6. Architectural trait extraction. (a) Main stem length estimation. A, B, and C represent the coordinate values at the highest, middle, and lowest points of the main stem point cloud along the z-axis, respectively. d_1 and d_2 denote the Euclidean distances between adjacent points. (b) Main stem diameter estimation. R denotes the radius of the red fitted circle on the XOY plane, with blue points representing the main stem projected onto this plane. (c) Internode distance estimation. The position of the nodes was represented by spheres, each assigned a different color. (d) Branch angle estimation. θ represents the angle between the branch and the main stem. $\overrightarrow{V_b}$ and $\overrightarrow{V_s}$ represent the direction vectors of the fitted lines for the branch and the main stem, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$\theta = \arccos\left(\frac{V_s \cdot V_b}{\|V_s\|\|V_b\|}\right) \tag{1}$$

The first step is to determine the entire main stem used for branch angle calculation. Due to the segmentation accuracy of PTv3, in a few cases, the segmented main stem may be incorrectly broken into several sections. To address this, DBSCAN ($\epsilon = 0.15$, MinPts $= 100$) was applied to cluster the segmented main stem points. If multiple clusters were formed, only the largest cluster (the one with the most points) was retained as the main stem.

Next, branch points near the main stem were selected. The Euclidean distance from the PTv3-segmented branch points to the main stem was calculated, and only branch points within the range of 0.5 cm to 4 cm were retained. Additionally, very small branches near the top of the plant were excluded. Specifically, any branches located within 20 cm vertically below the highest point of the plant were ignored.

Then, the selected branch points were clustered using DBSCAN ($\epsilon =$ 2 cm, MinPts $=$ 10) to obtain individual branch instances (the red points in Fig. 6d). Next, the main stem points near each branch were identified. The Euclidean distance from each individual branch to the main stem was calculated, and all points within 10 cm were retained as the corresponding main stem points for each branch (the blue points in Fig. 6d).

Finally, Principal Component Analysis (PCA) was performed separately on the branch and main stem point clouds, with the first principal component taken as the primary direction vector. The growth directions of the branch and main stem were used as the directions of the vectors. Since the branch and main stem share the same growth direction, branch angles greater than 100° were discarded during the calculation.

The individual branch instances were obtained in the previous step, and each branch corresponded to a node on the main stem (Fig. 6c). First, the lowest point of each branch (i.e., the point with the minimum z-coordinate) was determined. Using this point as a reference, a 1 mm-thick main stem point cloud slice was extracted downward along the z-axis. The point with the minimum z-coordinate within this slice was considered the node position. The difference in z-coordinates between two adjacent node positions was defined as the internode distance in this experiment.

2.4.2.3. Leaf trait. The total leaf volume of a cotton plant reflects the overall number of foliage and serves as an important trait of its photosynthetic capacity and biomass accumulation potential (Chen et al., 2017). A greater amount of foliage contributes to increased energy production, thereby supporting the formation and development of cotton bolls. In addition, total leaf volume is indicative of the plant's vegetative growth status and is closely associated with canopy structure, and water requirements (Gao et al., 2024). A voxel-based method was used to estimate the leaf volume of each plant. First, the 3D axis-aligned bounding box (AABB) of the segmented leaf point cloud was computed (Fig. 7). The bounding box was then divided into cubic voxels with a side length of $l = 1$ cm. Finally, the number of voxels n containing at least one leaf point was counted, and the total leaf volume V was estimated accordingly. Each voxel has a volume of $V_c = 1$ (cm3).

$$V = nV_c \tag{2}$$

The vertical distribution of foliage plays a crucial role in light interception and airflow within the canopy, reflecting the canopy structure and resource use efficiency. An optimal vertical leaf distribution helps coordinate light interception throughout the canopy, enhances light utilization by the middle and lower fruiting branches, and ultimately promotes uniform boll development (Blessing et al., 2020). Based on the last step method, the foliage of the entire plant have been segmented into voxels. The cotton plant was then vertically divided from bottom to top into layers with a thickness of 5 cm, the leaf volume within each layer was calculated, and a vertical leaf distribution map was generated.

2.5. Implementation details

The training of the point cloud deep learning models was conducted on the HiPerGator high-performance computing cluster. The

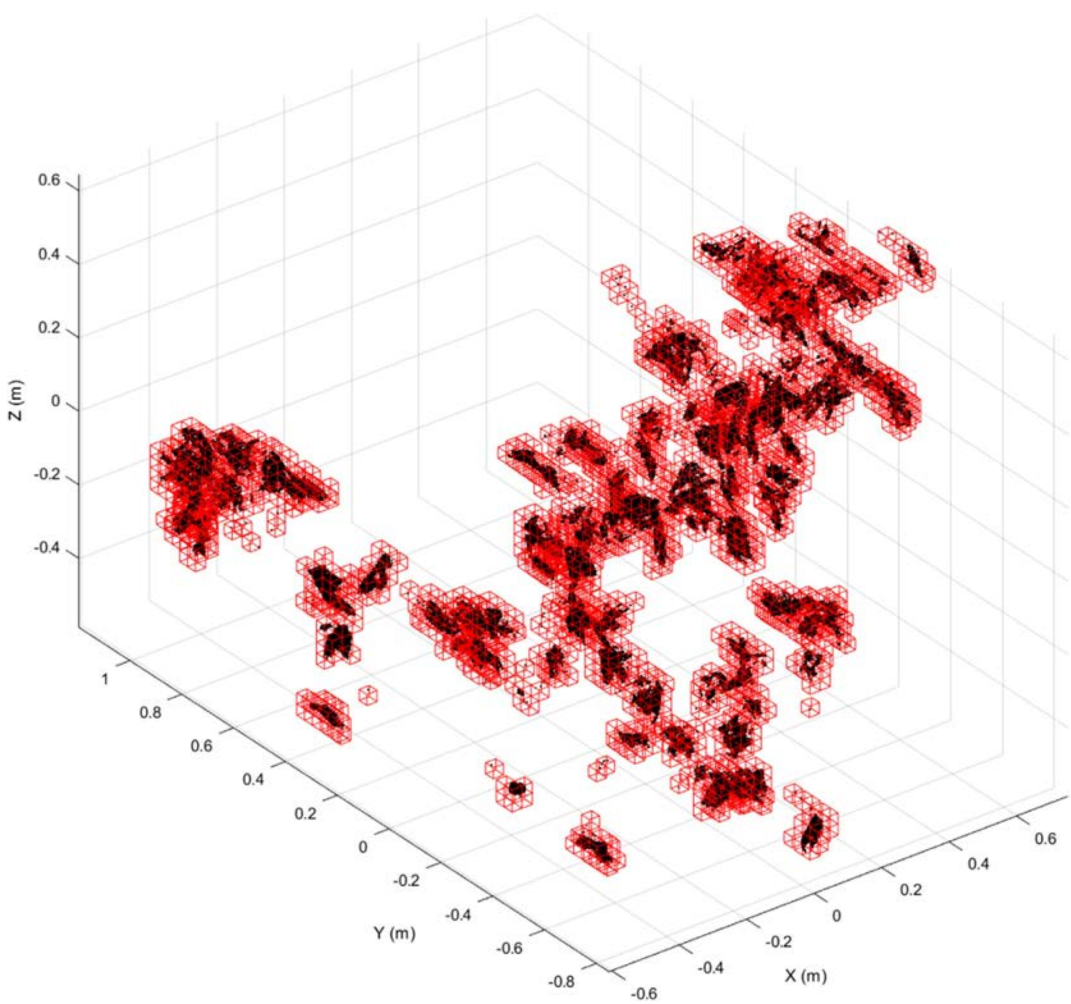

Fig. 7. Voxel-based leaf volume estimation. Red cubes represent voxels containing foliage points. The size of the cubes is set to 3 cm for visualization purposes, while a 1 cm size was used for actual volume estimation. Black dots indicate the segmented foliage points. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

CottonStruct-H dataset was trained on a node equipped with an NVIDIA A100 GPU (80 GB), whereas the CottonStruct-L dataset was trained on a separate node with an NVIDIA L4 GPU (24 GB). All training was performed under a Linux operating system. All models were subsequently tested and evaluated on the NVIDIA L4 GPU. Additionally, several hardware-related metrics—such as inference time and GPU memory footprint—were measured to assess computational efficiency. These evaluations were performed using the CottonStruct-L dataset with the batch size set to 1 during testing. The phenotypic trait extraction in this study was conducted on the CottonStruct-H dataset. The post-processing of cotton boll counting and the extraction of phenotypic traits were performed on a local computer, which has a 13th Gen Intel (R) Core (TM) i7-13700K processor at 3.40GHz and 32GB of RAM. The software used is MATLAB R2019b. More details can be found on GitHub: https://github.com/UGA-BSAIL/Cotton_plants_with_foliage.git.

2.6. Evaluation metrics

In this study, various point cloud segmentation metrics were derived from the confusion matrix, including Precision (P), Recall (R), F1 score, overall accuracy (OA), Intersection over Union (IoU), and mean Intersection over Union (mIoU). IoU measures the overlap between the predicted region and the ground truth for a specific class, while mIoU is the average IoU across all classes, providing an overall assessment of segmentation performance, where N is the total number of labels and i denotes the i-th label. Precision is the proportion of correctly predicted positive cases among all predicted positives, Recall is the proportion of correctly predicted positives among all actual positives, and F1 score is the harmonic mean of Precision and Recall. Overall Accuracy (OA) measures the proportion of correctly classified points among all points across all classes. Here, TP (true positive) refers to the number of correctly predicted positive cases, FP (false positive) refers to the number of incorrectly predicted positive cases, and FN (false negative) refers to the number of missed positive cases. Additionally, the evaluation of traits was conducted using root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). Here, N_i denotes the predicted value of the i-th sample, m_i denotes the corresponding ground truth, and n is the total number of samples. All phenotypic traits in this study were extracted and evaluated only on our self-collected dataset (CottonStruct-H), while CottonStruct-L was used solely for the evaluation of point cloud segmentation models.

Inference latency refers to the average time required for the model to process a single input sample during inference. It is typically measured in milliseconds (ms) per frame and reflects the model's responsiveness in practical applications. FPS indicates the number of input samples the model can process per second during inference and is calculated as the reciprocal of the inference latency. GPU memory footprint refers to the total amount of GPU memory consumed by the model during inference. It reflects the memory efficiency of the model and determines whether the model can be deployed on hardware with limited GPU resources.

$$IoU_i = \frac{TP_i}{TP_i + FP_i + FN_i} \tag{3}$$

$$mIoU = \frac{\sum_i^N IoU_i}{N} \tag{4}$$

$$P_i = \frac{TP_i}{TP_i + FP_i} \tag{5}$$

$$R_i = \frac{TP_i}{TP_i + FN_i} \tag{6}$$

$$F1\ score_i = \frac{2 \times P_i \times R_i}{P_i + R_i} \tag{7}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(N_i - m_i)^2} \tag{8}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|N_i - m_i| \tag{9}$$

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{N_i - m_i}{m_i}\right| \tag{10}$$

3. Results

3.1. Performance of point cloud semantic segmentation

PTv3 achieved the best performance in the two-class segmentation task (cotton bolls vs. non-cotton bolls) of cotton plants on the CottonStruct-H dataset. Compared to PointNet++, PointNeXt, and PTv2, PTv3 significantly improved the mIoU by 46.0%, 12.4%, and 6.7%, respectively. Other evaluation metrics also achieved the best results with PTv3, demonstrating its advantage in cotton plant point cloud segmentation. We present the mean results of the four-fold cross-validation in Table 3, and more detailed results are provided in Supplementary Table S1. The superior performance of PTv3 is largely due to its attention-based architecture, expanded receptive field, and ability to model long-range spatial dependencies, all of which are crucial for accurately segmenting complex and occluded biological structures such as cotton bolls. Notably, PointNet++ exhibited minimal segmentation capability for leafy cotton plants, with all evaluation metrics significantly lower across the four-fold cross-validation, indicating its limited segmentation capability under conditions of occlusion. Under four-fold cross-validation, PTv2 demonstrated superior segmentation performance compared to PointNeXt, with an average mIoU that was 5.7% higher. PTv3 consistently outperformed all baseline models across all evaluation metrics, demonstrating its superior and stable performance.

PTv3 also achieved the best performance in the two-class segmentation task (cotton bolls vs. non-cotton bolls) on the CottonStruct-L dataset, outperforming its results on the CottonStruct-H dataset (Table 4). In terms of mIoU, PTv3 surpassed PointNet++, PointNeXt, and PTv2 by 17.8%, 0.8%, and 0.2%, respectively, and demonstrated superior performance across other evaluation metrics. Comparative analysis on datasets with varying structural complexity highlights the limitations of PointNet++. Among all models, it showed the poorest performance under the same dataset size. Its architecture relies solely on local feature aggregation through set abstraction layers and lacks explicit modeling of spatial relationships between neighboring points, resulting in poor robustness under occlusion. On the CottonStruct-H dataset, PointNet++ exhibited almost no segmentation capability, and while its performance slightly improved on CottonStruct-L, the IoU for cotton bolls was still only 67.3%. These results indicate that PTv3 offers a clear

Table 3
Performance comparison of different models for boll and non-boll segmentation of cotton plants on CottonStruct-H dataset (the numbers in the table are in percentage %).

		P	R	F1 score	IoU	OA
PointNet++	Boll	49.9	18.0	23.8	14.0	
	Non-boll	83.1	95.5	88.7	79.8	
	Overall	66.5	56.7	56.3	46.9	80.4
PointNeXt	Boll	84.5	78.3	81.3	68.8	
	Non-boll	95.1	96.7	95.9	92.1	
	Overall	89.8	87.5	88.6	80.5	93.3
PTv2	Boll	**96.0**	80.1	87.3	77.5	
	Non-boll	95.6	**99.3**	97.4	95.0	
	Overall	95.8	89.7	92.3	86.2	95.8
PTv3	Boll	93.4	**94.2**	**93.8**	**88.4**	
	Non-boll	**98.7**	98.6	**98.7**	**97.4**	
	Overall	**96.1**	**96.4**	**96.2**	**92.9**	**97.8**

Table 4
Performance comparison of different models for boll and non-boll segmentation of cotton plants on CottonStruct-L dataset (the numbers in the table are in percentage %).

		P	R	F1 score	IoU	OA
PointNet++	Boll	87.9	74.2	80.4	67.3	
	Non-boll	94.0	97.5	95.8	91.8	
	Overall	91.0	85.9	88.1	79.6	93.0
PointNeXt	Boll	**98.2**	96.8	97.5	95.1	
	Non-boll	98.8	99.3	99.1	98.2	
	Overall	98.5	98.1	98.3	96.6	98.7
PTv2	Boll	**98.2**	97.2	97.7	95.5	
	Non-boll	99.3	**99.6**	99.4	98.9	
	Overall	98.7	98.4	98.6	97.2	99.1
PTv3	Boll	**98.2**	**97.6**	**97.9**	**95.9**	
	Non-boll	**99.4**	**99.6**	**99.5**	**99.0**	
	Overall	**98.8**	**98.6**	**98.7**	**97.4**	**99.2**

advantage in segmentation accuracy, particularly on structurally complex datasets. Despite the smaller size of the CottonStruct-H dataset, PTv3 consistently outperformed the other three models, confirming its robustness and strong generalization capability.

In the qualitative results, PTv3 showed the closest alignment with the ground truth (GT), outperforming other models, particularly on the more challenging samples from the CottonStruct-H dataset (Fig. 8). All Point Transformer models successfully segmented cotton bolls on plants with sparse foliage. However, for plants with dense structures (bottom row of Fig. 8), PTv3 achieved the most complete cotton boll segmentation, followed by PTv2. On the CottonStruct-L dataset, both PTv3 and PTv2 delivered highly accurate segmentation results (Fig. 9). In contrast, PointNeXt often misclassified points on the main stem as cotton bolls—a mistake not observed in the Point Transformer series. These results collectively demonstrate that PTv3 is the most suitable model for segmenting cotton plants with foliage in this study.

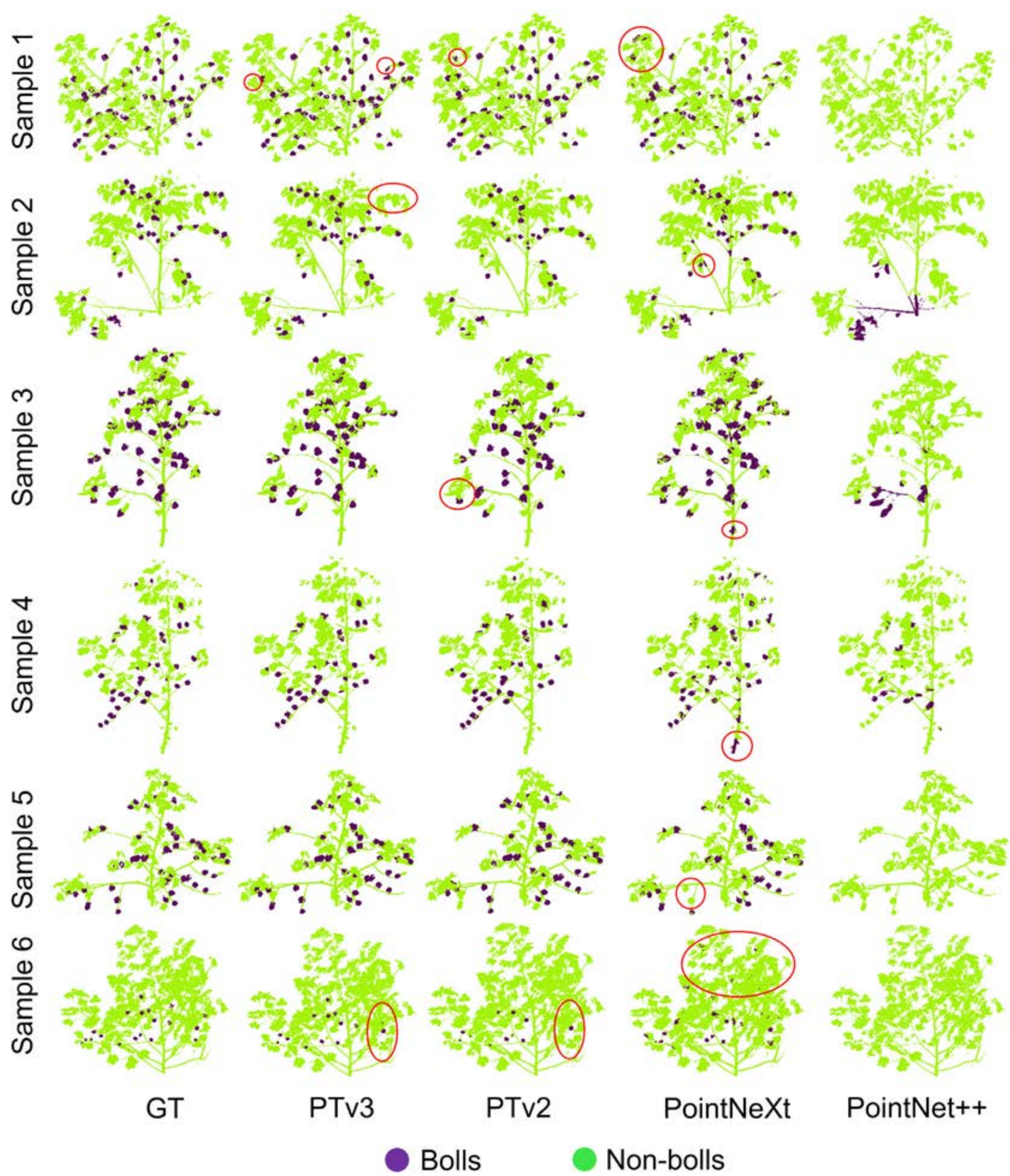

Fig. 8. Visualization of segmentation results of cotton plants with two-class (cotton bolls and non-cotton bolls) by different models on the CottonStruct-H dataset. Each row represents the segmentation results of the one sample under different models. GT stands for ground truth. The red circles highlight some segmentation errors, and PointNet++ shows no segmentation capability on this dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. The impact of label granularity on point cloud segmentation

As the number of segmentation classes increases, the mIoU tends to decrease because certain categories (e.g., branches or the main stem) account for only a small proportion of the points, leading to lower segmentation metrics. On the CottonStruct-H dataset, branches are heavily occluded by foliage, resulting in fragmented point cloud representations and severely affecting segmentation accuracy. Specifically, for the four-class classification, the average mIoU was only 84.5%, which was 8.4% lower than that of the two-class classification and 5.4% lower than that of the three-class classification. This decline was mainly attributed to the branch class, whose IoU was only 68.8% in the four-class segmentation. We present the mean results of the four-fold cross-validation in Table 5, and more detailed results are provided in Supplementary Table S2. On the CottonStruct-L dataset, all evaluation metrics were excellent. As shown in Table 6, the branch class was the only category with slightly lower segmentation accuracy, but it still reached 88.8%.

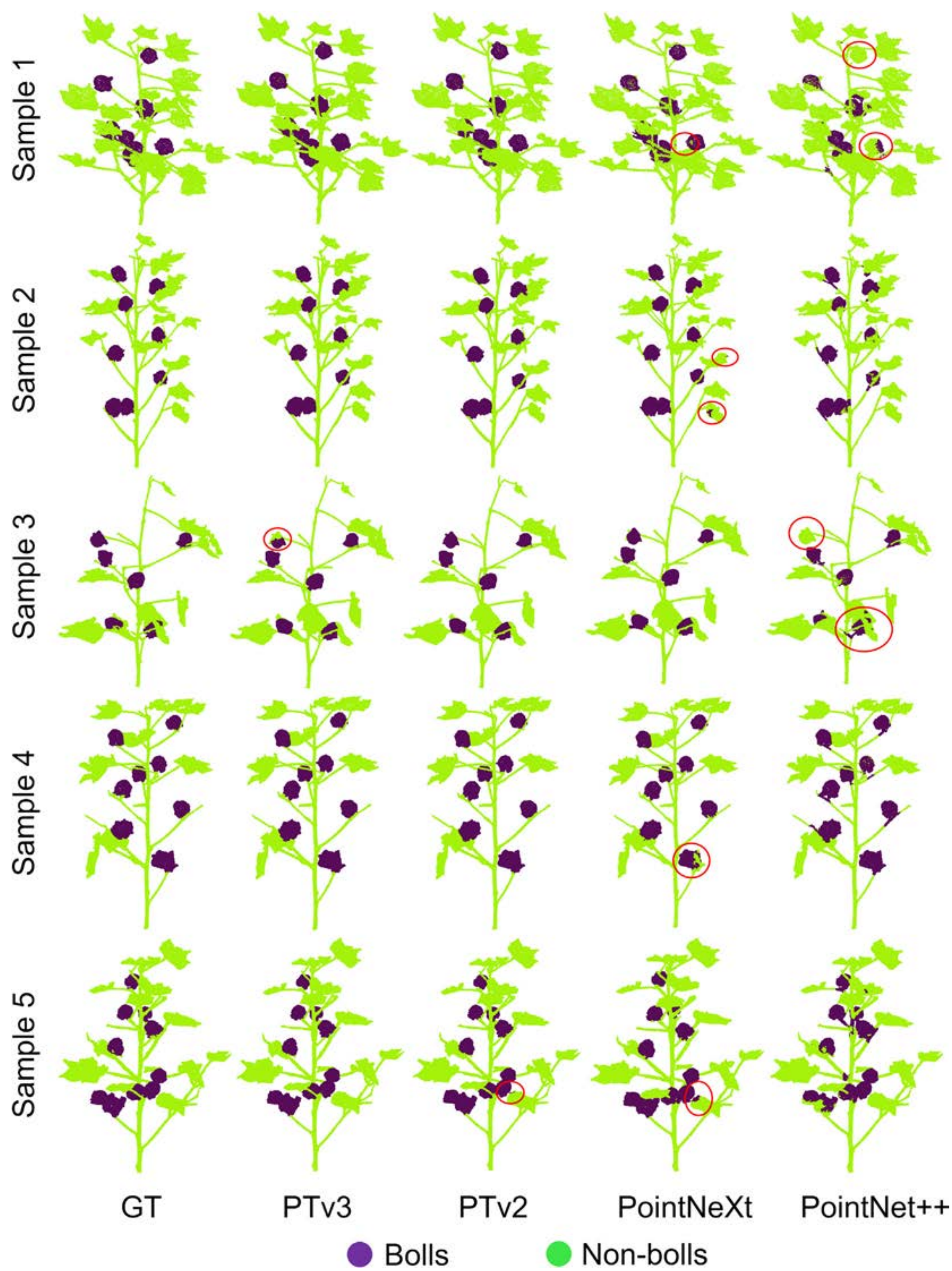

Fig. 9. Visualization of segmentation results of cotton plants with two-class (cotton bolls and non-cotton bolls) by different models on the CottonStruct-L dataset. Each row represents the segmentation results of the one sample under different models. GT stands for ground truth. The red circles highlight some errors for comparison with the GT. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Different segmentation granularities had only a minor impact on the segmentation results of cotton bolls. In the two-class segmentation on the CottonStruct-H dataset (boll IoU = 88.4%, the mean value of four-fold cross-validation), the segmentation accuracy of cotton bolls was slightly higher than that of the other segmentation schemes. In the three-class (boll IoU = 86.0%) and four-class (boll IoU = 86.6%) segmentations, the segmentation accuracy of cotton bolls was very similar. On the CottonStruct-L dataset, the IoU values for cotton boll segmentation in the two-class, three-class, and four-class schemes were 95.9%, 95.7%, and 95.2%, respectively. Although segmentation accuracy decreased slightly as the number of classes increased, this difference was negligible.

The visualization of segmentation results demonstrated that PTv3 effectively segmented all classes. In the three-class segmentation task on the CottonStruct-H dataset, the cotton boll predictions were highly consistent with the annotated ground truth (GT). The segmentation accuracy was particularly high for the lower part of the main stem, whereas errors were mainly observed at the top (Fig. 10a). This is attributed to the high point cloud density and thickness at the stem base, which facilitates precise segmentation. In contrast, the upper stem region is characterized by sparse points and heavy occlusion by leaves, resulting in reduced accuracy. Additionally, occasional misclassification of branches as main stem segments was observed (e.g., Sample 2 in Fig. 10a), although such cases were infrequent. For the three-class segmentation results on the CottonStruct-L dataset, the errors are not obvious (Fig. 10b), and PTv3 is fully capable of segmenting such structurally simple data.

In the four-class segmentation task on CottonStruct-H dataset, PTv3 successfully segmented branches and achieved accurate results at the junctions between the main stem and branches. However, the primary errors remained at the top of the main stem, where some regions were misclassified as branches. As shown in Sample 1 and Sample 5 of Fig. 11a, dense foliage at the stem apex resulted in sparse point cloud of the main stem, leading to reduced segmentation accuracy. The accuracy of branch segmentation was greatly affected by the quality of the point cloud. When the branch regions were captured at a higher density and were more clearly visible, PTv3 yielded better results. In addition, removing the foliage class from the visualization made the structural features of the plant more distinct (bottom row of Fig. 11a). Overall, aside from occasional segmentation errors at the stem apex, the segmentation performance across plant organs was satisfactory. A further limitation was observed in the form of point cloud incompleteness: extensive leaf occlusion hindered the device's ability to capture certain internal branch structures, particularly those located within the canopy interior. In the four-class segmentation on CottonStruct-L dataset, the results were almost identical to the ground truth, indicating that PTv3 is fully capable of segmenting such simple cotton plants.

PTv3 achieved the highest segmentation accuracy on the cotton datasets used in this study while maintaining comparable efficiency (Table 7). Although PointNet++ exhibited the highest inference efficiency, its segmentation performance was poor on both datasets and was therefore excluded from further consideration. PointNeXt is also a relatively efficient model and achieved high segmentation accuracy on the CottonStruct-L dataset, in which each sample contains 40,960 points. However, its performance degraded substantially on the more challenging CottonStruct-H dataset, indicating that PointNeXt is more suitable for datasets with lower structural complexity. Although PTv3 contains far more parameters than PTv2, its GPU memory usage and inference latency increased only slightly compared with PTv2 on the CottonStruct-L dataset, rather than exhibiting exponential growth. On the CottonStruct-H dataset, where each sample contains approximately 100,000 points, PTv2 showed a sharp increase in memory usage as the point cloud size grew, primarily due to the high memory cost of its neighborhood feature extraction. In contrast, PTv3 exhibited a much milder increase in memory consumption, demonstrating that its architectural optimization provides a better trade-off between segmentation accuracy and computational efficiency. In summary, segmentation accuracy, in addition to efficiency, is a primary objective of this study, and thus PTv3 was ultimately selected as the preferred model.

Table 5
Four-fold cross-validation of segmentation results of cotton plants labeled with three and four classes using PTv3 on the CottonStruct-H dataset (the numbers in the table are in percentage %).

		P	R	F1 score	IoU	OA
3-classes	Boll	**92.5**	92.4	92.4	86.0	
	Main stem	95.4	**91.9**	**93.7**	**89.6**	
	Foliage-branch	97.9	97.8	97.8	94.1	
	Overall	**95.2**	**94.1**	**94.7**	**89.9**	**96.7**
4-classes	Boll	90.6	**95.1**	**92.8**	**86.6**	
	Main stem	**97.0**	89.8	93.2	87.3	
	Branch	78.1	85.3	81.5	68.8	
	Foliage	98.2	96.9	97.5	95.2	
	Overall	91.0	91.8	91.3	84.5	95.9

Table 6
Comparison of PTv3 segmentation results for different classes of cotton plants on CottonStruct-L datasets (the numbers in the table are in percentage %).

		P	R	F1 score	IoU	OA
3-classes	Boll	98.0	**97.6**	**97.8**	**95.7**	
	Main stem	96.1	95.4	95.7	91.8	
	Foliage & branch	99.0	99.2	99.1	98.3	
	Overall	**97.7**	**97.4**	**97.6**	**95.3**	**98.7**
4-classes	Boll	**98.6**	96.5	97.5	95.2	
	Main stem	**96.9**	**97.3**	**97.1**	**94.4**	
	Branch	93.8	94.3	94.0	88.8	
	Foliage	99.3	99.8	99.6	99.1	
	Overall	97.2	97.0	97.1	94.3	98.4

3.3. Phenotypic traits

The cotton boll counting results obtained in this experiment demonstrated high accuracy and strong agreement with the ground truth obtained through manual counting (Fig. 12). Two approaches were evaluated: direct clustering using DBSCAN and an enhanced method that integrated DBSCAN with a post-processing step. As shown in Fig. 12, the predictions from DBSCAN combined with post-processing (red dots and red regression line) aligned more closely with the 1:1 reference line (dashed), indicating improved consistency with the true boll count. In contrast, results from the DBSCAN-only method (blue dots and blue line) exhibited a slight underestimation trend, as reflected by a lower slope in the regression eq. (0.88 vs. 0.96). From a quantitative perspective, the post-processing method reduced RMSE, MAE, and MAPE by 1.64, 1.30, and 2.7%, respectively. The enhanced performance confirms the effectiveness of the proposed post-processing strategy in refining clustering outcomes and eliminating under-segmented boll instances. By performing a secondary segmentation on clustered cotton bolls, the post-processing step significantly improved the accuracy of boll count estimation.

Through data visualization (Fig. 13), the spatial positions and distributions of cotton bolls at different heights were clearly observed and analyzed. Each cluster was assigned a unique color and an oriented bounding box (OBB) to represent an individual segmented cotton boll, with its spatial position within the entire plant clearly mapped. In addition, the bar charts of the three samples shown in the second row of Fig. 13 illustrate the number of cotton bolls within each 10 cm height interval, as well as the proportion of bolls distributed at different heights. Given that the number of cotton bolls per plant is one of the most critical factors determining yield (Hu et al., 2018), this comprehensive visualization provided valuable insights into the boll distribution and yield potential across the entire plant.

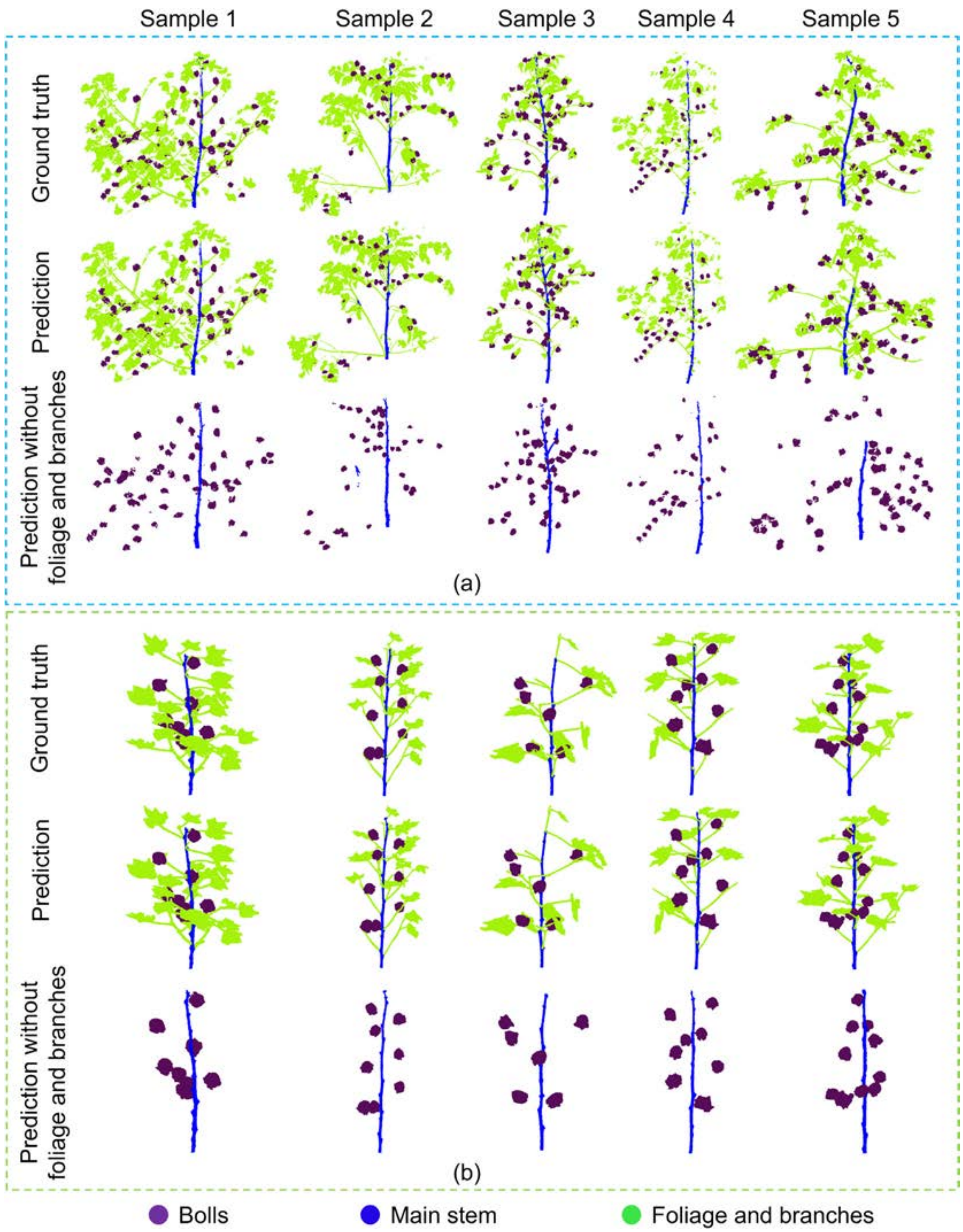

Fig. 10. Visualization of PTv3-based three-class segmentation results on CottonStruct-H (a) and CottonStruct-L (b) compared with the ground truth. Each column represents the same plant sample.

The spatial distribution of cotton bolls along the vertical direction revealed different patterns across various plants (Fig. 13). For example, the cotton bolls in Sample 1 and Sample 2 are mainly concentrated in the middle part of the plant, accounting for more than 50% of the total number of bolls. The number of bolls in the bottom part of the Sample 1 plant is slightly higher than that of the Sample 2 plant. These boll distribution patterns are likely influenced by multiple factors, including genotype, planting density, and light availability. In contrast, the Sample 3 plant exhibited a distinct top-heavy distribution, with a large number of bolls clustered at the upper canopy. This pattern reflects a growth tendency characterized by late-season boll formation at the plant apex, potentially due to delayed boll setting or enhanced apical dominance. Overall, these diverse distribution patterns highlight the variability in cotton plant architecture and reproductive allocation strategies, which can significantly affect yield formation.

By comprehensively evaluating the main stem length, main stem diameter, branch angle, and internode distance, all traits exhibited strong correlations between the predicted results and the ground truth measurements (Fig. 14). Specifically, the prediction accuracy for main stem length was the highest, with an average per-plant error of only 4.09%. The branch angle estimation ranked second, with an average error of approximately 7.04%. The accuracies for main stem diameter and internode distance were slightly lower, as these traits have smaller physical sizes and are thus more sensitive to measurement errors. For

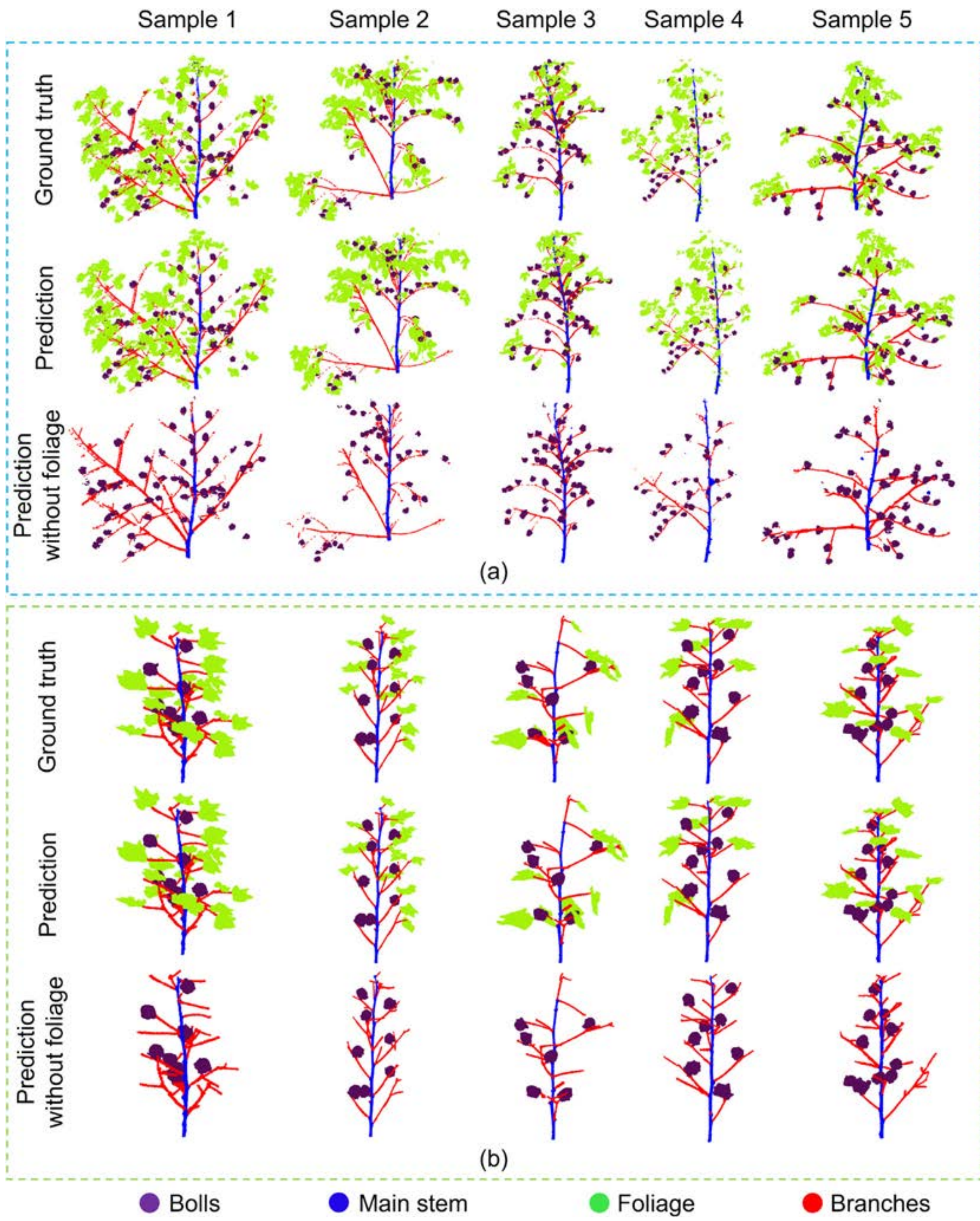

Fig. 11. Visualization of PTv3-based four-class segmentation results on CottonStruct-H (a) and CottonStruct-L (b) compared with the ground truth. Each column represents the same plant sample.

Table 7
Comparison of segmentation efficiency across different models on two datasets, where L denotes the CottonStruct-L dataset and H denotes the CottonStruct-H dataset.

Model	Parameters (M)	Inference Latency (ms)		GPU Memory (GB)		FPS	
		L	H	L	H	L	H
PointNet++	1.7	671	1607	2.9	4.9	1.49	0.62
PointNeXt	1.0	844	4071	0.9	1.9	1.18	0.25
PTv2	3.9	3491	22,421	2.5	13.8	0.29	0.04
PTv3	46.2	3732	20,563	3.1	4.2	0.27	0.05

example, when evaluating the main stem diameter, the fitted circle was more easily affected by noise, impacting the accuracy of diameter estimation. Overall, these results demonstrate the feasibility of extracting key structural traits of cotton plants using 3D point cloud data, laying a solid foundation for future architectural trait extraction of outdoor plants.

Significant differences in leaf volume were observed among different plants, even among those of the same genotype, reflecting the influence of both genetic factors and environmental conditions on canopy development (Fig. 15). These results highlight the substantial variability in canopy architecture across plant samples. Samples with larger leaf volumes may possess greater photosynthetic potential, contributing to enhanced biomass accumulation and yield, whereas plants with smaller

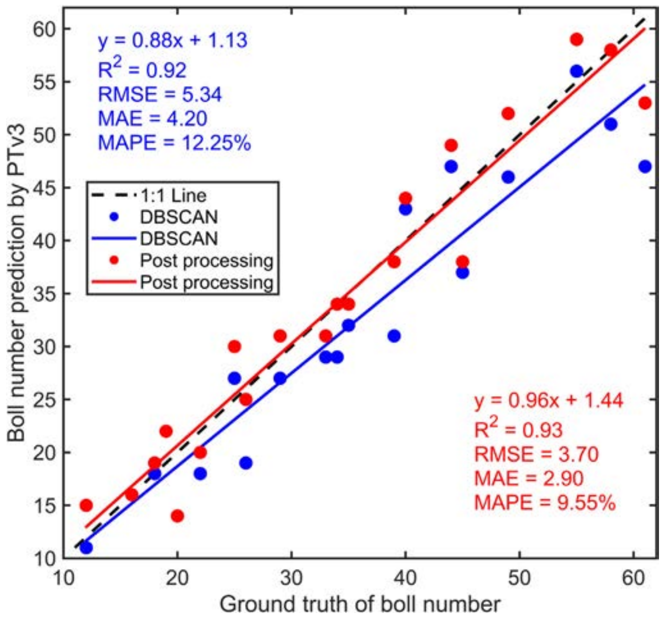

Fig. 12. Linear regression analyses of cotton boll number prediction from PTv3 and the ground truth in the CottonStruct-H dataset. The blue line represents the regression between the cotton boll count obtained using DBSCAN and the ground truth, while the red line represents the regression between the post-processed cotton boll count and the ground truth. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

leaf volumes may exhibit better adaptation to resource-limited or stress-prone environments.

Mapping the distribution of foliage along the plant height for three different cotton plants revealed significant differences in vertical leaf allocation patterns (Fig. 16). In the leftmost plant, leaf volume was primarily concentrated in the upper canopy region, and the overall leaf density was relatively sparse. In the middle plant, the leaf volume was distributed more evenly across the canopy, although slight accumulation was observed at mid-height levels. In contrast, the rightmost plant exhibited a relatively larger leaf volume at lower heights, which gradually decreased toward the top. These results highlight the differences in canopy architecture and spatial leaf distribution among individual cotton plants, reflecting potential variations in growth vigor, branching structure, and light interception strategies.

4. Discussion

This study proposes a transformer-based point cloud segmentation model (PTv3) for segmenting the point clouds of cotton plants with foliage, enabling accurate cotton boll counting and architectural traits extraction. This method provides breeders and physiologists with a convenient and non-destructive tool for identifying genotypes with desirable phenotypic traits. The focus of this experiment was on cotton plants with foliage, which are substantially more challenging to segment due to the dense and irregular distribution of leaves. In contrast to cotton plants without foliage, where stems and bolls are clearly exposed, the overlapping and occluding leaves in leafy plants significantly hinder accurate segmentation. The segmentation performance difference is evident in our experiments: PointNet++ achieved a high mIoU of 90.2% for cotton plants without foliage (Jiang et al., 2025), but

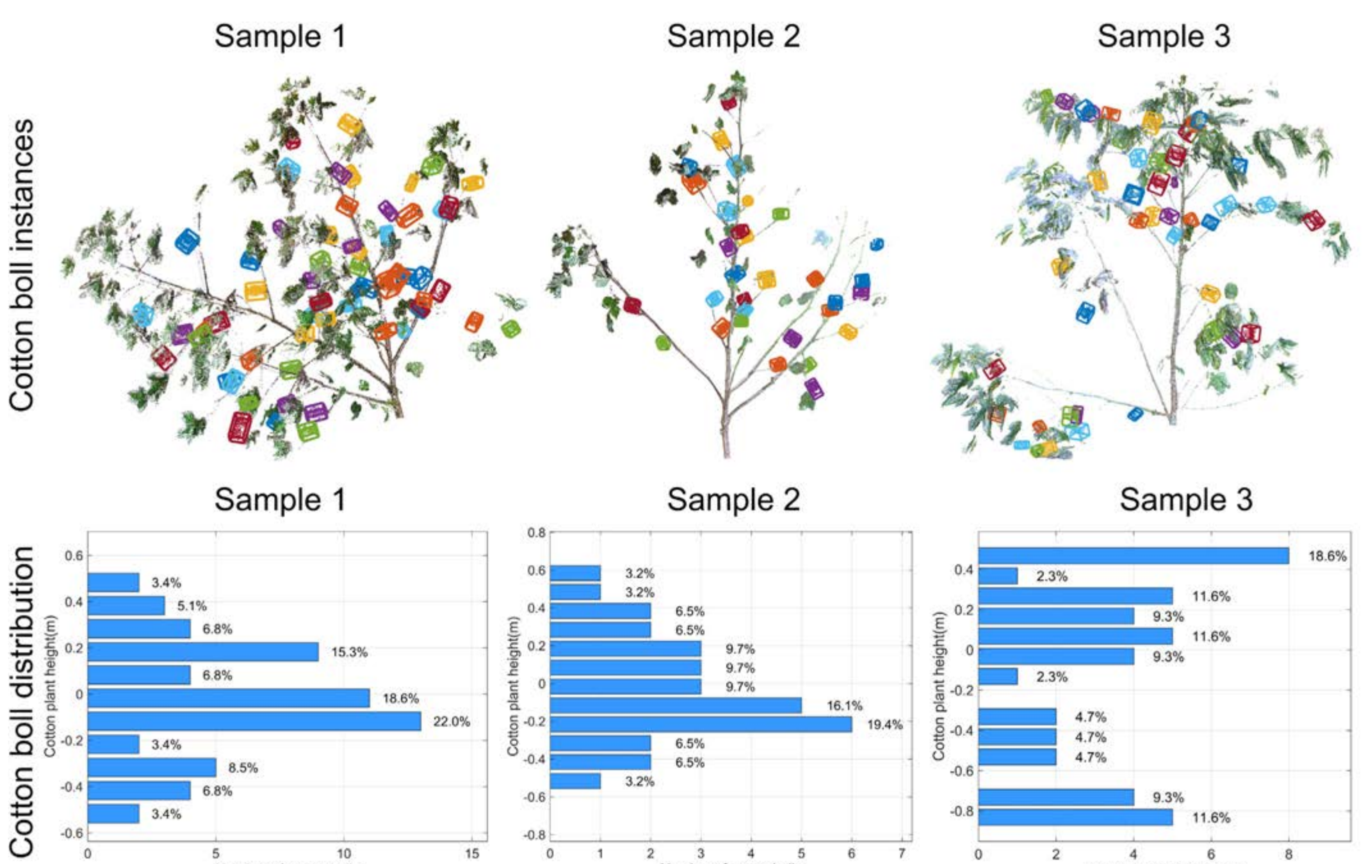

Fig. 13. The cotton boll segmentation under the two-class classification was performed using PTv3 in the CottonStruct-H dataset. The first row shows the instance segmentation results of cotton bolls, and the second row presents the distribution map of the cotton boll height (z-axis) across different levels of the plant.

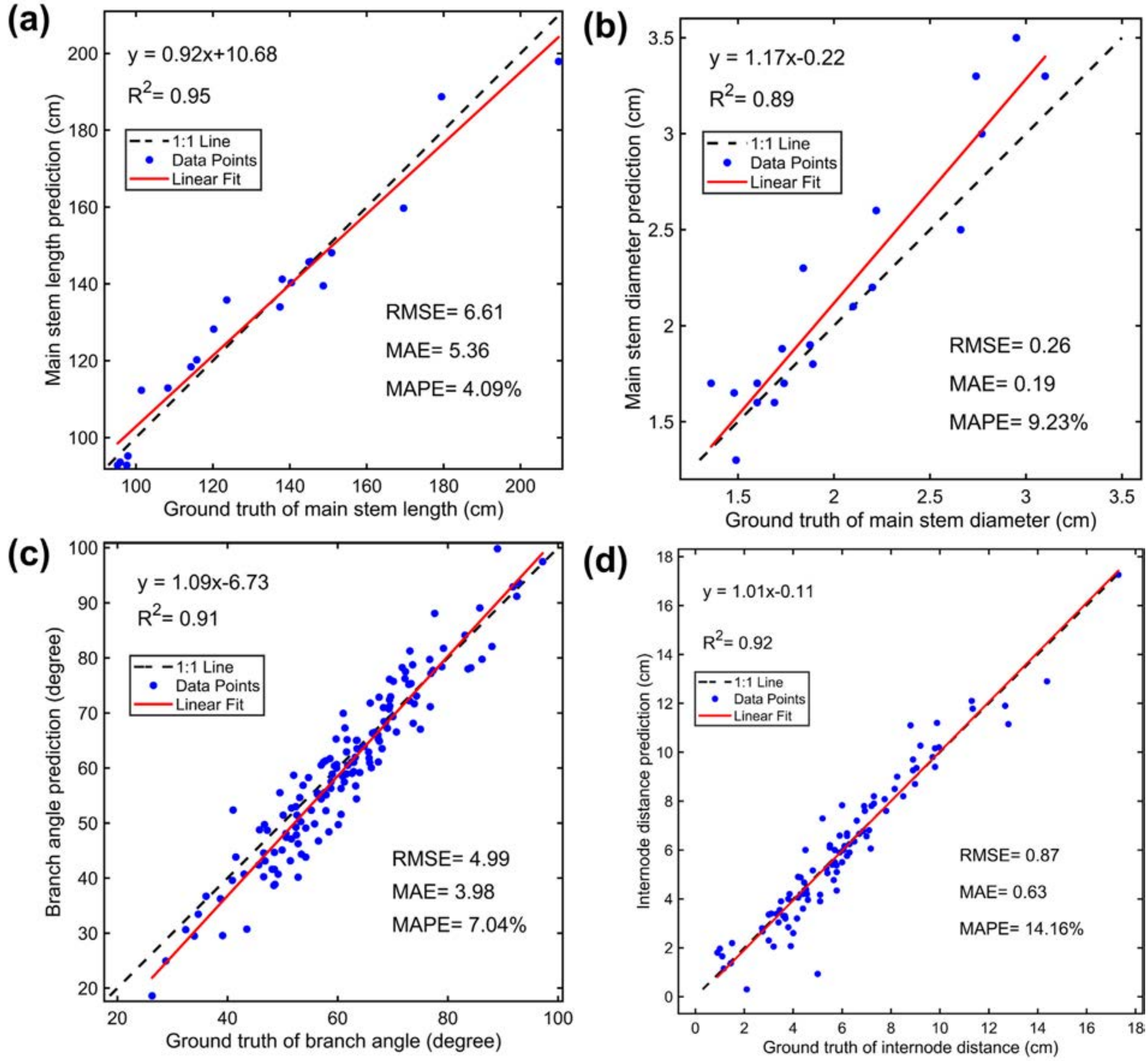

Fig. 14. Linear regression analyses of four plant architectural traits between the predicted values by our workflow and the ground truth in the CottonStruct-H dataset. (a) Main stem length. (b) Main stem diameter. (c) Branch angle. (d) Internode distance.

the performance dropped to 46.9% for plants with foliage. This stark decline reveals the limitations of traditional point-based models such as PointNet++, which rely heavily on local geometric features and struggle to model contextual information needed to resolve occlusions. Previous studies have also shown that PointNet++ performs poorly on challenging data (van Marrewijk et al., 2025). For instance, in segmentation studies of point clouds from power transmission corridors, PointNet++ achieved an mIoU of only 51.1% (Wang et al., 2023). In contrast, the PTv3 model we employed fully exploits the global modeling capability of the transformer architecture, making it particularly well-suited for segmenting complex cotton plants with dense foliage and demonstrating strong generalization ability. Consistent with our findings, recent studies have also reported that incorporating transformer architectures into segmentation models significantly improves performance on plant point cloud datasets (Liu et al., 2025).

PTv3 exhibited superior segmentation results for our cotton plants with foliage. Compared with the segmentation results of PEPNet on cotton seedlings, our two-class segmentation (cotton bolls and non-bolls) on CottonStruct-H improved the mIoU by 1.6%. When segmenting cotton plants into three classes (cotton bolls, main stems, and foliage-branches), the mIoU decreased by only 1.4%. These results indicate that PTv3 achieved a comparable segmentation accuracy to PEPNet, even on the more complex cotton plants with foliage (Yan et al., 2024). Moreover, on CottonStruct-L, the two-class and three-class segmentation results were higher by 6% and 4%, respectively. PointsegAt is a stem-leaf segmentation model for cotton seedlings designed based on PointNet++. Its segmentation performance is very close to ours. However, the structure of cotton seedlings is much simpler than that of the mature cotton plants with foliage used in our study (Hao et al., 2024). In our previous research, PVCNN achieved mIoUs of 96.2%, 89.8%, and 81.4% for the segmentation of cotton bolls, main stems, and branches of cotton plants without foliage (Saeed et al., 2023). These values were 9.6%, 2.5%, and 12.6% higher than the mIoUs obtained from the four-class segmentation on CottonStruct-H. Compared with the four-class segmentation results on CottonStruct-L, the mIoU for cotton bolls was 1% higher, while those for main stems and branches were 4.6% and 7.4% lower, respectively. Our branch segmentation performed the worst, as the branches were heavily surrounded by foliage, making their point cloud very sparse. Many branches were fragmented and discontinuous, which made them difficult to distinguish.

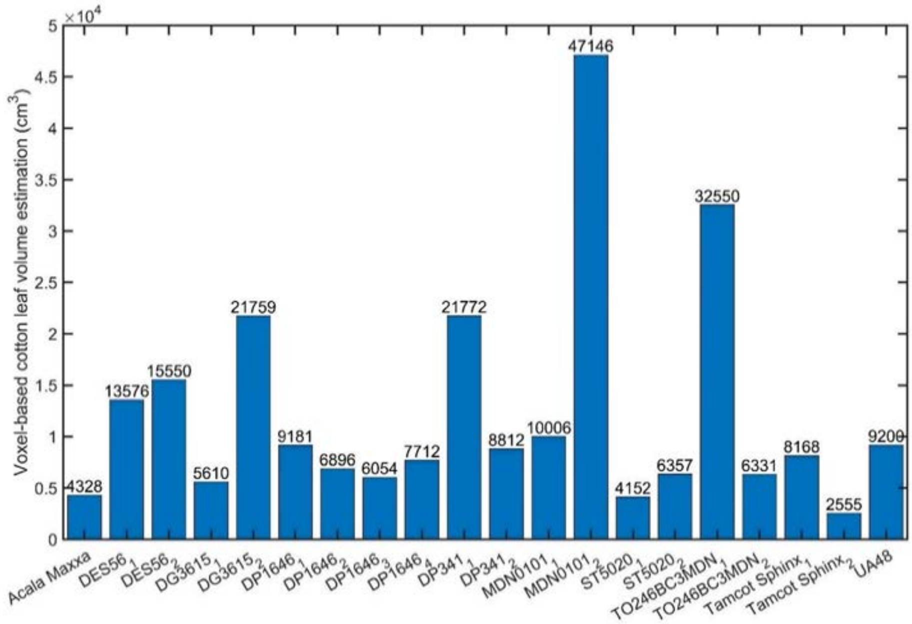

Fig. 15. Estimation of leaf volume across different plant samples in the CottonStruct-H dataset.

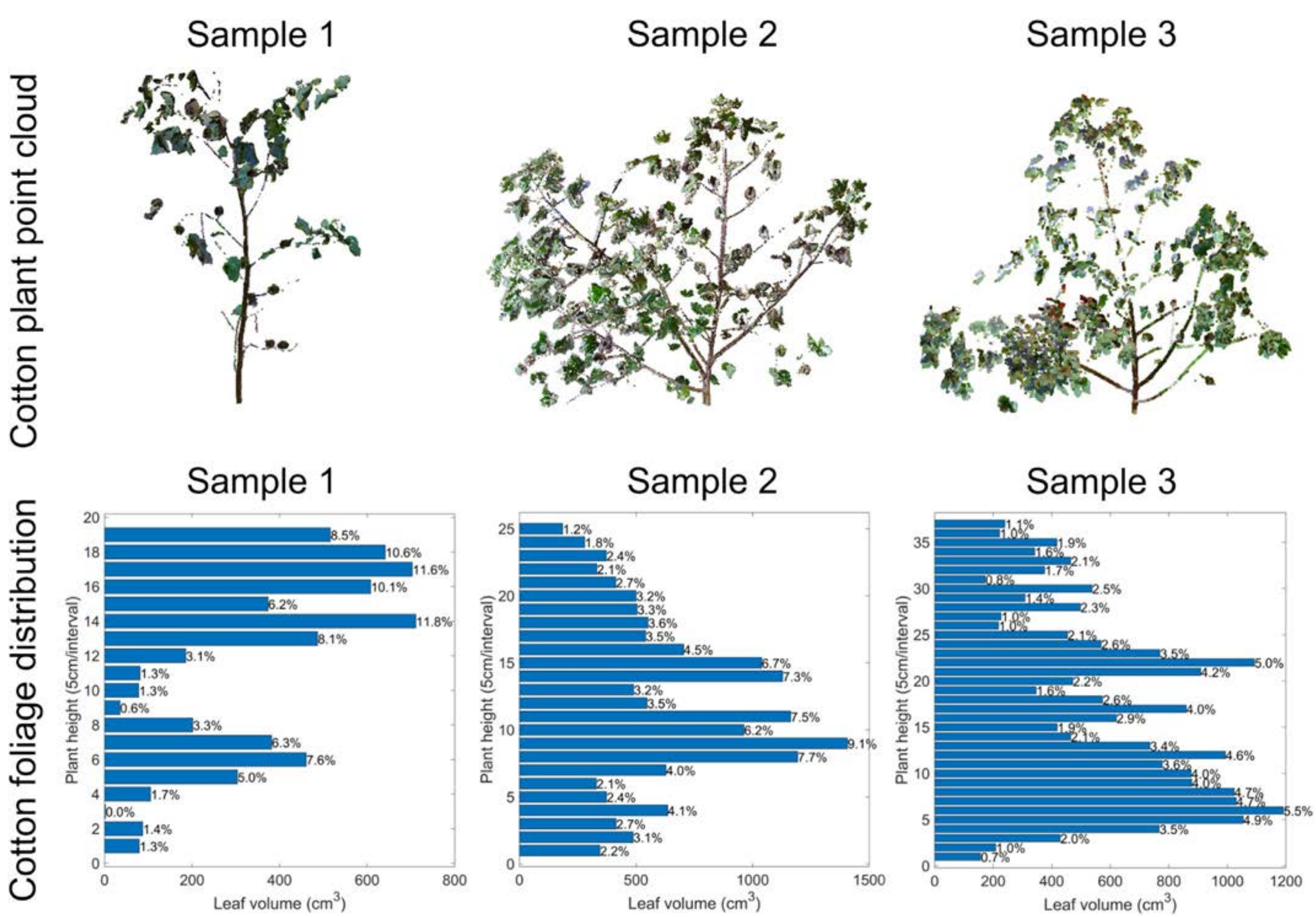

Fig. 16. The distribution of leaf volume at different heights of the plant in the CottonStruct-H dataset. The first row displays the point clouds of individual cotton plants, while the second row illustrates corresponding vertical distribution of leaf volume.

Comparative analysis on the CottonStruct-H and CottonStruct-L cotton plant datasets shows that PTv3 is the most suitable model for cotton plant segmentation in this study, as it achieves the highest segmentation accuracy, which is the primary focus of our research, followed by PTv2. PointNeXt and PointNet++ produced inferior segmentation results, particularly on the CottonStruct-H dataset. PTv3 not only provides high segmentation accuracy for cotton plants but also significantly reduces memory consumption and improves processing speed, as detailed in the PTv3 paper (Wu et al., 2024). For example, on the ScanNet dataset, using a single RTX 4090 GPU with a batch size of 1, PTv3 achieves a training latency approximately half that of PTv2 and an inference latency about one-third as large. PTv3 also greatly reduces GPU memory consumption (Table 7). Overall, PTv3 achieves a favorable trade-off between accuracy and efficiency.

The DBSCAN threshold was determined through multiple manual tests and is related to point cloud density and boll distribution. For the CottonStruct-H dataset, we started with smaller thresholds. The key parameter of DBSCAN is the neighborhood radius. We evaluated four neighborhood radii (1.0 cm, 2.0 cm, 3.5 cm, and 5.0 cm) to identify the optimal value (Fig. 17). The results showed that a radius of 3.5 cm achieved the most accurate boll counting with a MAPE of 9.55%. Considering that a MAPE below 10% meets our breeding accuracy requirement, 2.0 cm (MAPE = 9.68%) is also acceptable. In this study, we selected 3.5 cm as it achieved the lowest MAPE.

Accurate segmentation of individual cotton plants enables the extraction of detailed organ-level phenotypic traits, such as boll count and spatial distribution, leaf area, and branching structure. These traits provide valuable indicators for assessing plant growth status and identifying developmental differences at various growth stages. For breeding experts, organ-level phenotypic information from single plants is crucial for understanding genotype performance and trait heritability (Großkinsky et al., 2015). By enabling quantitative comparison of different cultivars, the extracted phenotypic data serve as a robust basis for varietal evaluation and selection, thereby accelerating the breeding of superior cultivars (Danilevicz et al., 2025). Moreover, such high-resolution phenotypic information lays the foundation for data-driven breeding strategies, integrating advanced phenotyping with genetic improvement to enhance cotton yield and quality.

There are a few limitations in this study. The cotton plant data used in this experiment was collected in an indoor environment, where lighting conditions and the background were relatively controllable. In contrast, outdoor environments present challenges for data collection due to variations in lighting, wind speed, and complex backgrounds. These factors may interfere with data collection and lead to missing or noisy plant point clouds. Currently, many studies cultivate cotton plants in pots within greenhouse environments and collect seedling point clouds indoors (Yan et al., 2024; Shen et al., 2024; Hao et al., 2024). Although some research efforts have focused on acquiring cotton point clouds in outdoor environments, these studies have primarily aimed at analyzing fruit-related traits, with limited attention given to architectural traits (Bolouri et al., 2024). This is largely because organs such as branches are relatively slender and challenging to capture with high precision in outdoor conditions. In addition, our study demonstrated that PTv3 achieved the best segmentation results on both the CottonStruct-H and CottonStruct-L datasets. However, the CottonStruct-H dataset has a relatively small sample size and limited genotypic diversity. As a next step, we plan to collect more cotton plants with different genotypes to further verify whether the segmentation accuracy of PTv3 can be significantly improved, which requires additional validation. In future work, we plan to design dedicated data acquisition systems and collect high-quality, plot-level cotton plant point clouds in field environments, enabling a more detailed analysis of architectural traits.

Furthermore, the terrestrial LiDAR (FARO Focus S70) used in this experiment is expensive, with a cost exceeding $20,000, making it difficult to popularize. This highlights the need to explore more cost-effective alternatives, such as using low-cost RGB-D cameras for point cloud acquisition or reconstructing plant 3D models from RGB images. However, a major limitation is that the point clouds captured by RGB-D cameras are relatively low quality. While they are sufficient for representing thicker main stems, they may suffer from missing data regarding finer structures such as branches. Neural Radiance Fields (NeRF, Mildenhall et al., 2021) and 3D Gaussian splatting are emerging high-quality 3D reconstruction methods, which have great potential and represent a promising direction for future research.

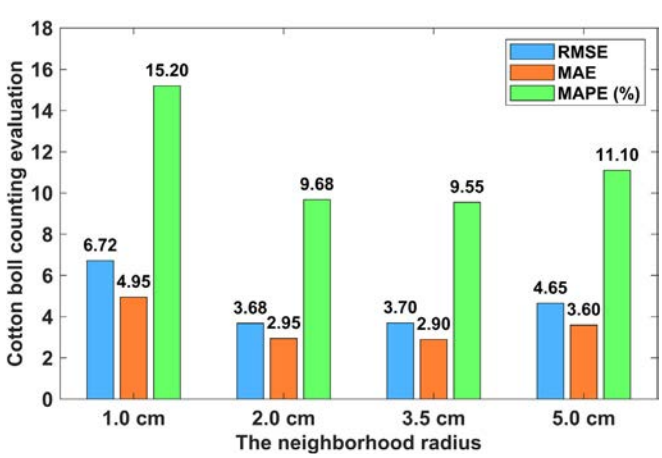

Fig. 17. Effect of different neighborhood radius thresholds on cotton boll counting. Four thresholds (1.0 cm, 2.0 cm, 3.5 cm, and 5.0 cm) were compared.

5. Conclusions

This study introduces a state-of-the-art transformer-based point cloud segmentation method for segmenting individual cotton plants and extracting architectural traits at a relatively early growth stage before defoliation. PTv3 was found to perform better than PointNet++, PointNeXt, and PTv2 in segmenting leafy cotton plants on two datasets with different levels of challenge, and it effectively enabled segmentation at varying class-granularities. The post-processing methods designed in this study achieved high accuracy in cotton boll counting and architectural traits were effectively extracted from segmented organs such as branches, main stems, and foliage. This approach provides an effective pathway for 3D plant phenotyping research and consequently facilitates cotton breeding. Potential limitations include the high cost of point cloud data collection and the limited number of samples analyzed. Future work will focus on reducing data collection costs, expanding data acquisition to outdoor environments, and extracting key architectural traits of cotton plants, including branch angles, internode distance, and main stem length.

CRediT authorship contribution statement

Lizhi Jiang: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Javier Rodriguez-Sanchez:** Writing – review & editing, Supervision, Data curation. **Peng W. Chee:** Writing – review & editing, Supervision, Resources. **Changying Li:** Writing – review & editing, Supervision, Software, Resources, Methodology, Funding acquisition. **Longsheng Fu:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aiia.2026.07.006.

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